Descriptive Statistics Topic Test 1

1. State whether each type of data is categorical nominal (N), categorical ordinal (O), numerical discrete (D), numerical continuous (C).
   a. Weight of truck loads   b. Nationality   c. Scores on a maths exam

a). Numeral Continuous (C)
b). Categorical Nominal (N)
c). Numeral discrete (D)

2. The number of people ordering pizzas each night:

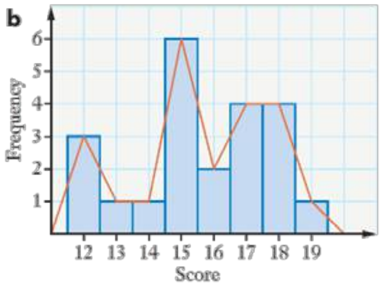
   15, 12, 17, 18, 18 ,15, 16, 13, 15, 17, 18, 12, 17, 14, 16, 15, 17, 18, 19, 15, 15, 12

   a. Draw a frequency distribution table.
   b. Draw a histogram and frequency polygon.
   c. Find the highest and lowest scores.
   d. Find the most frequent score.

a

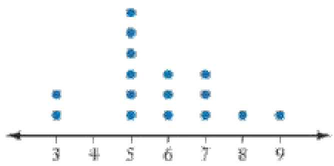
Pizzas	Frequency
12	3
13	1
14	1
15	6
16	2
17	4
18	4
19	1

b



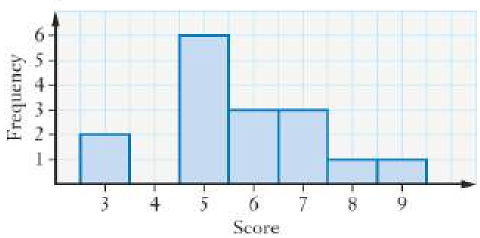
c Highest 19, lowest 12   **d** 15

3. Lauren surveyed her friends and had them rank a film from 1 to 10. The dot plot shows the results of her survey.
   a. How many friends did Lauren survey?
   b. What was the most common ranking?
   c. Draw the results in a histogram.
   d. What percentage of rankings were above 4?
   e. What fraction of Lauren's friends ranked the film below 4?



a 16   **b** 5

c



d). $\frac{14}{16} \times 100 = \boxed{87.5\%}$

e). $\frac{2}{16} = \boxed{\frac{1}{8}}$

4. The number of students absent from Year 12 for the past 9 days was as follows: 19, 20, 15, 14, 31, 17, 20, 16, 22
   a. What is the mean? Answer correct to one decimal place.
   b. Find the interquartile range.
   c. Is 31 an outlier for this set of data? Justify your answer with calculations.

2(a)
Using the statistic mode on the calculator.
$\bar{x} = 19.3333\ldots \approx 19.3$
2(b)
Using the statistic mode on the calculator.
$Q_1 = 15.5$ and $Q_3 = 21$
$\text{IQR} = Q_3 - Q_1 = 21 - 15.5 = 5.5$

2(c)
Outlier
$$\begin{aligned}\text{Upper limit} &= Q_3 + 1.5\times\text{IQR} \\ &= 21 + 1.5\times 5.5 \\ &= 29.25\end{aligned}$$
$\therefore$ 31 is an outlier as it is above the upper limit of 29.25.

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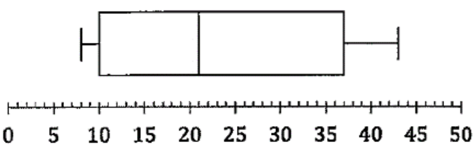
5. Class A has 28 students and achieved a mean on an assessment task of 72.5%. Class B has 32 students and achieved a mean on the same assessment task of 78.5%. What was the mean mark for both classes? Answer correct to one decimal place.

Class A total number of marks
$72.5 \times 28 = 2030$
Class B total number of marks
$78.5 \times 32 = 2512$
$$\text{Mean} = \frac{2030 + 2512}{28 + 32} = 75.7\%$$

6. Sixteen students travelling to Brisbane have their bags weighted at the airport.
   a. The five-number summary for the weights of the bags was: $8, 10, 21, 37, 43$.
      Construct an accurate box-and-whisker plot to display the distribution of the weights of the bags.
   b. How many students had bags weighing less than 10?

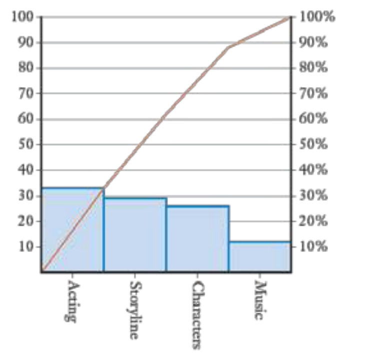
6(a)
Minimum = 8, Q_1 = 10, Median = 21,
Q_3 = 37, Maximum = 43



6(b)
$Q_1 = 10$
(25% of the data is less than Q_1)
Number of students $= 0.25 \times 16 = 4$
$\therefore$ 4 students had bags weighing less than 10.

7. A survey into why people like a movie is shown below. Draw a Pareto chart for the data set.

Reason	Votes
Acting	33
Storyline	29
Music	12
Characters	26



8. The table shows the results of a survey to find the distance people must travel to work.
   a. Find the modal class.
   b. Find the mean.
   c. What is the median?
   d. Find the standard deviation and variance.

Distance (km)	Frequency
0–4	6
5–9	8
10–14	4
15–19	7
20–24	3

a). 8 is the most frequent score
so the modal class is
$\boxed{5-9}$

b). Distance (km) Class centre
0 - 4 2
5 - 9 7
10 - 14 12
15 - 19 17
20 - 24 22

$$\text{Mean} = \frac{2\times6 + 7\times8 + 12\times4 + 17\times7 + 22\times3}{6+8+4+7+3} = \boxed{10.75}$$

c). Total score = 28
Middle score is between the 14^{th} and 15^{th} score.

$$\text{Average} = \frac{7+12}{2} = \boxed{9.5}$$

d). Standard deviation $= \boxed{6.6}$
Variance $= \boxed{43.97}$

9. Nutritionists tested the fat content of beef sausages and chicken sausages. Each sausage tested was given a 'fat content' score – the higher the score, the higher the fat content of the sausage. The results of the tests are shown below in the stem-and-leaf plot.

Beef	Stem	Chicken
	0	3 4 7
1	1	0 1 1 3
9 7 6 5	2	5 6 7 8
9 8 7 5 5 4 4 0	3	1 2 2 3 6
8 7	4	5
5 5 5 3 2 1	5	

a. What is the highest score in the test?

b. What is the mode score for beef sausages?

c. How many chicken sausages were tested.

d. Compare the types of sausages in terms of centre and spread.

8(a)
Highest score in the test was 55 for beef sausages. (Biggest number)

8(b)
Mode is beef sausages is 55.
(Score that occurs the most number of times)

8(c)
There were 17 chicken sausages tested.

8(d)

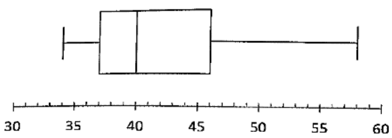
Beef	Stem	Chicken
	0	3 4 7
1	1	0 1 1 3
7 6 5	2	5 6 7 8
9 8 7 5 5 4 4 0	3	1 2 2 3 6
8 7	4	5
5 5 5 3 2 1	5	

Beef sausages:
Median: 37.5
Range: $55 - 11 = 44$
IQR: $52 - 30 = 22$

Chicken sausages:
Median: 26
Range: $45 - 3 = 42$
IQR: $32 - 10.5 = 21.5$

$\therefore$ Beef sausages have a higher median fat content (centre is higher). The median for beef sausages is 37.5 and chicken sausages is 26. The spread in the fat content is about the same for beef and chicken sausages. Beef sausages have a range of 44 (IQR is 22) and chicken sausages have a range of 42 (IQR is 21.5).

10. The box-and-whisker plot shows the assessment results of 160 students.



Which of the following statements is false?

A. Data is negatively skewed.

B. Median score is 40.

C. 120 students achieved a score greater than the lower quartile.

D. 40 students achieved a score greater than 46.

14 (A)
(A) Data is more on the left side. Data is positively skewed. False.
(B) Median is 40 (middle line of the box). True.
(C) 75% of 160 $= 0.75 \times 160 = 120$ True.
(D) 25% of 160 $= 0.25 \times 160 = 40$ True.

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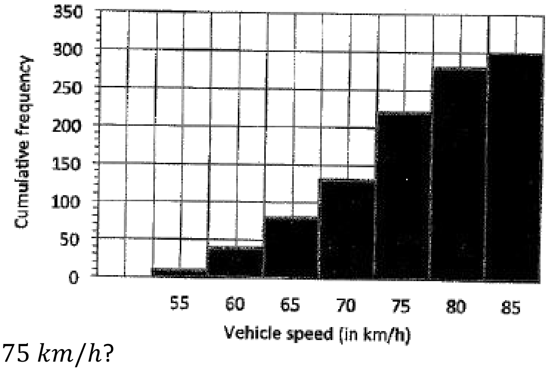
11. The speeds of a vehicle are shown below in the cumulative frequency histogram.



a. What is the frequency of 75 km/h?

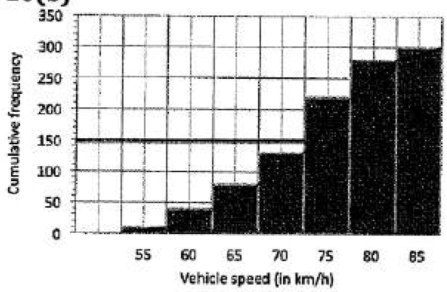
b. Construct a cumulative frequency polygon (or ogive) on this graph.

c. Use the graph to estimate the median.

16(a)

Frequency of $75 = 220 - 130$
$= 90$

16(b)



16(c)

There are 300 vehicles, so the median is the speed of the 150th vehicle.

Median is approximately 74 km/h (see the graph in 16(b)).

12. Find the mean, mode and median of this data: $5, 5, 4, 4, 10, 8, 9, 10, 6, 7, 7, 3, 4$

12). 3, 4, 4, 4, 5, 5, 6, 7, 7, 8, 9, 10, 10

Mean = 6.31

Mode = 4

Median = 6

13. Which of the statements about the data below is correct?

$$7, 8, 10, 13, 13, 15$$

A. The range is greater than the mode.
B. The median is greater than the mode.
C. The mean is greater than the range.
D. 50% of the scores are greater than the mode.

13). 7, 8, 10, 13, 13, 15

Range = 15 − 7 = 8

Mean = 11

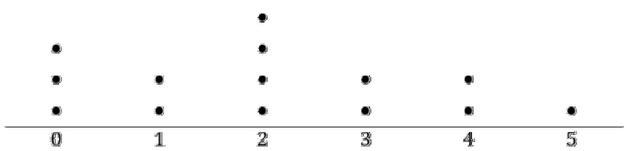
Mode = 13

Median = $\frac{10+13}{2}$ = 11.5

A. ✗ C. ✓

B. ✗ D. ✗

14. A sample of 14 people were asked to indicate the time (in hours) they had spent watching television on the previous night. The results are displayed in the dot plot below. What is the mean and sample standard deviation of these times? Give your answers correct to one decimal place.



$\bar{x}$ = 2.1

s.d. = 1.6

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15. Twenty students were asked how many hours of study they completed per week. What is the mean number of hours of study completed by the students per week?

Hours per week	Frequency
0–4	5
5–9	10
10–14	3
15–19	2

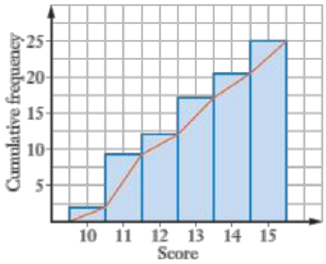
Class cent	Frequency
2	5
7	10
12	3
17	2

$\bar{x} = \frac{2\times5 + 7\times10 + 12\times3 + 17\times2}{5+10+3+2}$

$\boxed{\bar{x} = 7.5}$

16. For the dot plot:
    a. Draw a cumulative frequency polygon.
    b. Find:
        i. the 1st quartile
        ii. the 3rd quartile
        iii. the 35th percentile
        iv. the 7th decile
        v. the 1st decile

a



b). (i). Total number of scores = 25

$25\% \times 25 = 6.25^{th}$ score

6.25^{th} score = $\boxed{11}$

(ii). $75\% \times 25 = 18.75^{th}$ score

18.75^{th} score = $\boxed{14}$

(iii). $35\% \times 25 = 8.75^{th}$ score

8.75^{th} score = $\boxed{12}$

(iv). $\frac{7}{10} \times 25 = 17.5^{th}$ score

17.5^{th} score = $\boxed{13.5}$

(v). $\frac{1}{10} \times 25 = 2.5^{th}$ score

2.5^{th} score = $\boxed{10.5}$

17. For this set of data, find:
    a. The mean
    b. The mode
    c. The median
    d. The range
    e. The interquartile range

Rain (mm)	Frequency
5	4
6	7
7	8
8	3
9	3

a). Mean = $\boxed{6.76}$

b). Mode = $\boxed{7}$

c). Total scores = 25

Middle = $\frac{25}{2} = 12.5$

$= 13^{th}$ score

Median = $\boxed{7}$

d). Range = $9 - 5 = \boxed{4}$

e). $Q_1 = \frac{6+6}{2}$

$= 6$

$Q_3 = \frac{7+8}{2}$

$= 7.5$

IQR $= 7.5 - 6$

$= \boxed{1.5}$

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18. Jane recorded the number of crocodile sightings each week in a region of the Northern Territory over several weeks.

$$5, 8, 9, 4, 11, 7, 9, 15, 17, 10, 8, 5, 9, 12$$

a. What was the mean number of crocodiles sighted per week?
b. Find the standard deviation.
c. Calculate the variance.

a). Mean = 9.2

b). Standard deviation = 3.5

c). Variance = 12.6

19. This table shows the results of an assessment task.

a. Find:

i. The mean   ii. The median   iii. The mode

b. Describe the shape of the distribution.

Score	Frequency
4	3
5	5
6	6
7	9
8	6
9	5
10	3

a). (i). Mean = 7

(ii). Median = 7

(iii). Mode = 7

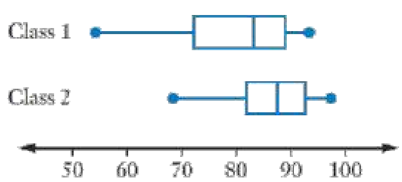
b). Symmetrical (normal distribution)

20. Mrs Spell's piano students earned the following marks in their piano examinations:
Class 1: 91, 86, 74, 92, 85, 89, 63, 71, 80, 91, 85, 72, 54, 78
Class 2: 97, 87, 69, 91, 88, 89, 93, 94, 71, 79, 84, 85, 88

a. Sketch parallel box plots showing this information.
b. For each class, find:

i. The median   ii. The interquartile range
iii. The mean   iv. The range

a



Class 1 : 54, 63, 71, 72, 74, 78, 80 | 85, 85, 86, 89, 91, 91, 92

Class 2 : 69, 71, 79, 84, 85, 87, $\boxed{88}$ 88, 89, 91, 93, 94, 97

b). (i). Median:

Class 1 : $\boxed{82.5}$

Class 2 : $\boxed{88}$

(ii). Class 1 : $Q_1 = 72$, $Q_3 = 89$

IQR = $\boxed{17}$

Class 2 : $Q_1 = \frac{79+84}{2} = 81.5$

$Q_3 = \frac{91+93}{2} = 92$

IQR = $\boxed{10.5}$

(iii). Mean:

Class 1 : $\boxed{79.4}$

Class 2 : $\boxed{85.8}$

(iv). Range:

Class 1 : $92 - 54 = \boxed{38}$

Class 2 : $97 - 69 = \boxed{28}$